

Building the Big Data Team

SIVA PRIYA M S



Agenda

- The Data Scientist
- The Team Challenge
- Different Teams, Different Goals
- Don't Forget the Data
- Challenges Remain
- Teams versus Culture
- Gauging Success





The Data Scientist

Role & Responsibilities

- Acts as the core analytical expert who extracts actionable insights from large, complex, and heterogeneous datasets.
- Combines **statistical modeling**, **machine learning**, **data engineering**, and **domain knowledge** to design predictive and prescriptive solutions.
- Builds analytical pipelines that include data acquisition, cleaning, feature engineering, model development, validation, and deployment.
- Collaborates with business units to translate ambiguous business problems into analytical formulations.

Key Skill Areas

- **Mathematics & Statistics:** probability models, optimization, regression, time-series, hypothesis testing.
- **Machine Learning & AI:** supervised/unsupervised algorithms, deep learning, recommendation systems.
- **Programming:** extensive use of Python/Scala/R; Apache Spark for distributed analytics.
- **Data Handling:** SQL/NoSQL systems, data lakes, unstructured data processing.
- **Communication:** ability to explain complex models to non-technical stakeholders.

The Team Challenge

Why Big Data Teams Are Hard to Build



Effective delivery techniques

- The skillset required is multidisciplinary;
- finding professionals with both analytical depth and engineering strength is difficult.
- Traditional IT teams are optimized for deterministic systems, not for iterative model development.
- Organizations often underestimate the resources required for data acquisition, storage, and cleansing.
- There is a gap between management's expectations and the real-world timeline needed for modeling cycles.

Cross-Functional Dependency

Effective work requires tight coordination between:

- Data engineers
- Data scientists
- Domain specialists
- DevOps/ML Ops engineers
- Business units

Typical Bottlenecks

1. Data access restrictions
2. Lack of standardized data pipelines
3. Fragmented infrastructure
4. Misalignment between analytical output and business needs

Different Teams, Different Goals

Team Structure Depends on Organizational Priorities

1. Technology-centric Enterprises

1. Focus heavily on scalable data platforms, streaming analytics, and automation.
2. Teams emphasize **engineering and real-time processing**.

2. Business-centric Enterprises

1. Emphasize dashboarding, reporting, marketing analytics, customer profiling.
2. Teams are oriented toward **descriptive and diagnostic insights**.

3. Research-oriented Organizations

1. Invest in experimentation, algorithm development, and advanced statistical models.
2. Teams focus on **innovation and long-term analytics capability**



Don't Forget the Data

Data Is the Foundation

- Analytics quality is directly limited by the **accuracy, completeness, and timeliness** of data.
- Without clean, discoverable, and well-managed datasets, even skilled teams underperform.

Key Data Elements

- Data governance: policies for lineage, privacy, standards, and lifecycle management.
- Metadata management: cataloging, schema documentation, and provenance tracking.
- Data quality controls: validation rules, anomaly detection, cleansing workflows.
- Data accessibility: secure yet flexible access models for analytics users.

Challenges Remain

Data-Driven Challenges

- **Data silos:** isolated datasets across departments.
- **Inconsistent formats:** need for normalization and schema harmonization.
- **Velocity issues:** difficulty handling real-time or near-real-time streams.

Challenges Remain



Technology Challenges

- Integration of legacy systems with modern big data platforms.
- Lack of scalable storage and compute infrastructure.
- Complexity of maintaining distributed frameworks like Hadoop/Spark.

Challenges Remain

People & Process Challenges

- Shortage of skilled professionals in ML Ops, cloud engineering, and data science.
- Poorly defined KPIs for analytics projects.
- Resistance from employees accustomed to traditional processes.

Gauging Success

Operational Metrics

Analytical Metrics

Organizational Metrics

Long-Term View